

IT486: Blockchains and Cryptocurrencies (3-0-0-3)

Autumn 2025-26

Course Placement

Open to students in the final year of the B.Tech and M.Tech programs.

Pre-requisites: You should know the basics of cryptography.

Instructor: Anish Mathuria

Objectives: This course explores the fundamentals of blockchains and cryptocurrencies.

Course topics

Topics covered include blind signatures, anonymous electronic cash, decentralized cryptocurrencies, proof of work, Bitcoin blockchain, bitcoin transactions, Script language, micropayments, lotteries, cryptocurrency exchange, proof of solvency, atomic currency swap, mining, mining pools, chain splits, ECDSA/Schnorr signatures, signature aggregation, Pedersen commitment, confidential transactions, simplified payment verification, hierarchical key generation, unlinkable transactions, coin mixing, ring signatures, transaction malleability, smart contracts, and Ethereum.

Course materials

We will use selected papers and references available online. The lecture slides will be made available in the Lecture folder as the course progresses.

The following references are useful.

- *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*, Narayanan, et. al. Princeton, 2016. ISBN: 978-0691171692.
- *Programming Bitcoin*, Jimmy Song, O'Reilly, 2019. ISBN: 978-93-5213-808-1.

Evaluation (tentative)

- 20% pop quizzes (in-lecture)
- 40% two in-sem exams
- 20% assignments (written + programming)
- 20% term paper

Course Outcomes

Upon completion of the course students should be able:

- to understand properties of blockchains and their applications
- to understand the fundamental principles and algorithms underlying bitcoin
- to understand the limitations of cryptocurrencies such as bitcoin

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X		X			X				